



INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSIC ACADEMY

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Course Code: SBT-DF203

Course Title: Basic Computer Skills for Digital Forensics

Lab: Lab 6 — Firewall Traffic Control and Forensic Verification

Operating System: Kali Linux

Laboratory Workspace: SBT-DF203_Lab6_2025_FWSD_11470_Peter_Oloo

Submission Date: 18 Sept 2026

Part 1 — Create Lab 6 workspace

```
(kali㉿kali)-[~]  
$ mkdir -p ~/SBT-DF203_Lab6_Peter_Oloo/{evidence,pcap,hashes,screenshots,logs,report,working}  
cd ~/SBT-DF203_Lab6_Peter_Oloo  
pwd  
/home/kali/SBT-DF203_Lab6_Peter_Oloo  
  
(kali㉿kali)-[~/SBT-DF203_Lab6_Peter_Oloo]  
$
```

2. Check the network before changing anything

```

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:96:2f:18 brd ff:ff:ff:ff:ff:ff
    inet 192.168.249.131/24 brd 192.168.249.255 scope global dynamic noprefixroute eth0
        valid_lft 1640sec preferred_lft 1640sec
    inet6 fe80::28b1:ff7a:7f30:eb46/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:9a:8c:20:bd brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ ip route
default via 192.168.249.2 dev eth0 proto dhcp src 192.168.249.131 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.249.0/24 dev eth0 proto kernel scope link src 192.168.249.131 metric 100

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ ip neigh
192.168.249.2 dev eth0 lladdr 00:50:56:fc:c1:5c REACHABLE

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$

```

3. Identify the Apache HTTP listener

```

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sudo ss -lntp | grep ':80'
[sudo] password for kali:

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sudo systemctl status apache2 --no-pager
o apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; disabled; preset: disabled)
   Active: inactive (dead)
   Docs: https://httpd.apache.org/docs/2.4/

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$

```

4. Record the original firewall ruleset

```

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sudo iptables-save > evidence/iptables_original.rules

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sudo iptables -L -n -v --line-numbers > evidence/iptables_original_verbose.txt

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sudo iptables -S > evidence/iptables_original_commands.txt

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sha256sum evidence/iptables_original.rules | tee hashes/iptables_original.sha256
c655c85fda249b6587e16ce2bf8ead9bbc8b05f73e4da7f8db96651b2382336d evidence/iptables_original.rules

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$

```

Capture the allowed HTTP traffic

```
(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ ip route
default via 192.168.249.2 dev eth0 proto dhcp src 192.168.249.131 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.249.0/24 dev eth0 proto kernel scope link src 192.168.249.131 metric 100

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$
```

```
(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ ls -lh pcap/allowed_http.pcapng
-rw-r--r-- 1 tcpdump tcpdump 24 Sep 18 15:24 pcap/allowed_http.pcapng

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sha256sum pcap/allowed_http.pcapng | tee hashes/allowed_http.sha256
704e5e5b3234433c01fcfd1b20a306e77e985038120492dc53965c3edd38a4ea  pcap/allowed_http.pcapng

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$
```

Insert the narrow DROP rule

```
(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sudo iptables -I INPUT -s 192.168.249.131 -p tcp --dport 80 -j DROP

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$
```

Verify the DROP rule

```
(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sudo iptables -L INPUT -n -v --line-numbers
Chain INPUT (policy ACCEPT 3 packets, 228 bytes)
num  pkts bytes target    prot opt in     out     source               destination           tcp dpt:80
1      0      0 DROP      tcp  --  *      *      192.168.249.131      0.0.0.0/0              tcp dpt:80

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$
```

```
(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sudo iptables -S INPUT
-P INPUT ACCEPT
-A INPUT -s 192.168.249.131/32 -p tcp -m tcp --dport 80 -j DROP

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$
```

Capture the blocked traffic

```

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sudo tcpdump -i eth0 -nn -s 0 -w pcap/blocked_http.pcapng 'tcp port 80'
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C0 packets captured
0 packets received by filter
0 packets dropped by kernel

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ ls -lh pcap/blocked_http.pcapng
-rw-r--r-- 1 tcpdump tcpdump 24 Sep 18 15:42 pcap/blocked_http.pcapng

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sha256sum pcap/blocked_http.pcapng | tee hashes/blocked_http.sha256
704e5e5b3234433c01fcfd1b20a306e77e985038120492dc53965c3edd38a4ea  pcap/blocked_http.pcapng

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$

```

Examine the firewall counters

```

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sudo iptables -L INPUT -n -v --line-numbers
Chain INPUT (policy ACCEPT 15 packets, 1392 bytes)
num  pkts bytes target    prot opt in     out     source               destination
1      0      0 DROP      tcp  --  *      *       192.168.249.131      0.0.0.0/0            tcp dpt:80

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$

```

Compare allowed and blocked captures

```

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ tcpdump -nn -r pcap/allowed_http.pcapng
reading from file pcap/allowed_http.pcapng, link-type EN10MB (Ethernet), snapshot length 262144

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ tcpdump -nn -r pcap/blocked_http.pcapng
reading from file pcap/blocked_http.pcapng, link-type EN10MB (Ethernet), snapshot length 262144

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ tcpdump -nn -tttt -r pcap/allowed_http.pcapng
reading from file pcap/allowed_http.pcapng, link-type EN10MB (Ethernet), snapshot length 262144

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ tcpdump -nn -tttt -r pcap/blocked_http.pcapng
reading from file pcap/blocked_http.pcapng, link-type EN10MB (Ethernet), snapshot length 262144

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$

```

Remove the exact rule

```

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sudo iptables -L INPUT -n -v --line-numbers
Chain INPUT (policy ACCEPT 23 packets, 2000 bytes)
num  pkts bytes target    prot opt in     out     source               destination
1      0      0 DROP      tcp  --  *      *       192.168.249.131      0.0.0.0/0            tcp dpt:80

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sudo iptables -D INPUT 1

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sudo iptables -L INPUT -n -v --line-numbers
Chain INPUT (policy ACCEPT 24 packets, 2076 bytes)
num  pkts bytes target    prot opt in     out     source               destination

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sudo iptables -S INPUT
-P INPUT ACCEPT

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$

```

Hash all important evidence

```
(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sha256sum pcap/*.pcapng | tee hashes/pcap_hashes.sha256
704e5e5b3234433c01fcfd1b20a306e77e985038120492dc53965c3edd38a4ea  pcap/allowed_http.pcapng
704e5e5b3234433c01fcfd1b20a306e77e985038120492dc53965c3edd38a4ea  pcap/blocked_http.pcapng

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ sha256sum evidence/* > hashes/evidence_sha256.txt

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$ cat hashes/pcap_hashes.sha256
704e5e5b3234433c01fcfd1b20a306e77e985038120492dc53965c3edd38a4ea  pcap/allowed_http.pcapng
704e5e5b3234433c01fcfd1b20a306e77e985038120492dc53965c3edd38a4ea  pcap/blocked_http.pcapng

(kali@kali)-[~/SBT-DF203_Lab6_Peter_Oloo]
$
```

Report Summary

This practical examined the behavior of a Linux host based firewall using iptables in an isolated laboratory environment. The goal was to find out if a source-specific firewall rule could be used to prevent HTTP traffic from an authorized laboratory client while still allowing normal HTTP connectivity before and after the controlled firewall modification.

The original network configuration, Apache HTTP listener and iptables ruleset were captured prior to making any changes to the firewall. The original firewall rules were exported and saved as evidence. A SHA-256 hash was calculated to support evidence integrity. Then a baseline HTTP request from the authorized laboratory client was performed to establish normal connectivity.

Capture of HTTP allowed packet before the change in the firewall. The capture was saved and hashed. The traffic observed was used as a baseline for comparison to the blocked connection later.

We then added a narrow iptables INPUT rule that would DROP TCP traffic to port 80 from the specified laboratory client IP address. The rule was verified by verbose iptables output, and its packet and byte counters were observed. A second attempt to make an HTTP connection from the authorized client was made and a separate packet capture was collected.

Overall, the practical demonstrated forensic value of correlation between firewall configuration and packet level evidence. We had a number of sources of evidence to determine the impact of the controlled firewall modification, including the state of the iptables rules, packet/byte counters, HTTP connection behavior, packet captures and SHA-256 hashes. All testing was conducted in the approved isolated

laboratory environment. The original ruleset and captured evidence were maintained for reporting purposes.